

Enterprise Server Deployment and Operating System Installation

Installation Guide

ITEP 414 – System Administration and Maintenance | Week 3 Portfolio Project

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Project Overview

This guide documents the deployment of Ubuntu Server for ABC Startup Solutions' first Linux server. The server is intended to support file sharing, remote administration, database hosting, web hosting, and internal services. It covers the full process from installation through verification, along with supporting research on boot technologies and operating system selection.

Objectives

- Install and configure Ubuntu Server following company standards.
- Enable secure remote administration using SSH.
- Verify server functionality after installation.
- Compare BIOS and UEFI firmware and document the Linux boot process.
- Compare Ubuntu Server, Windows Server, and Rocky Linux for enterprise use.
- Produce documentation that another administrator can follow without additional guidance.

Software Requirements

- Oracle VirtualBox or VMware Workstation
- Ubuntu Server LTS ISO (latest stable version)
- Windows Server Evaluation ISO (bring-home activity)

Virtual Machine Specifications

Component	Configured Value
Name	Ubuntu-Server-Week03
RAM	4 GB
CPU	2 Virtual Processors
Storage	40 GB (VDI/VMDK)
Network	NAT
Optical Drive	Ubuntu Server ISO

Step-by-Step Installation Procedure

Language and Keyboard

Selected English as the installation language and English (US) as the keyboard layout.

Network Configuration

Accepted DHCP for automatic IP assignment. Assigned IP address:

10.0.2.15

Hostname

Hostname assigned: server01

User Account

Created a non-root administrative user for day-to-day management, avoiding direct use of the root account for security best practice.

Field	Value
Full Name	Andrei Villanueva
Username	kokokqt
Password	

Disk Partitioning

Selected Guided – Use Entire Disk for a straightforward, single-disk layout suitable for a lab/startup environment.

Detail	Value
File system	[e.g., ext4]
Disk size	40 GB

SSH Server

Enabled 'Install OpenSSH Server' during setup so the server can be administered remotely immediately after first boot.

Additional Packages

No additional packages were installed, per project instructions, to keep the base image minimal and secure.

Completing Installation

Restarted the virtual machine and removed the installation ISO from the virtual optical drive when prompted.

Configuration Details

Summary of the finalized server configuration:

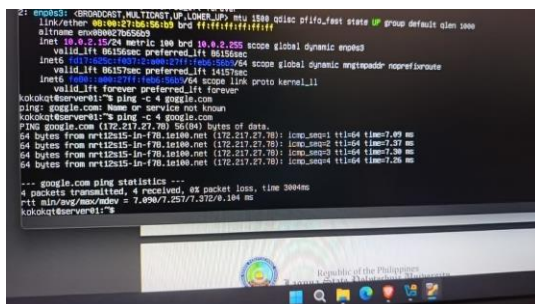
Setting	Value
Hostname	server01
Primary user	Villanueva
Network mode	NAT (DHCP)
SSH	Enabled (OpenSSH Server)
Firewall	[document ufw status if configured]

Verification Commands and Results

Logged in successfully using the created administrative account.

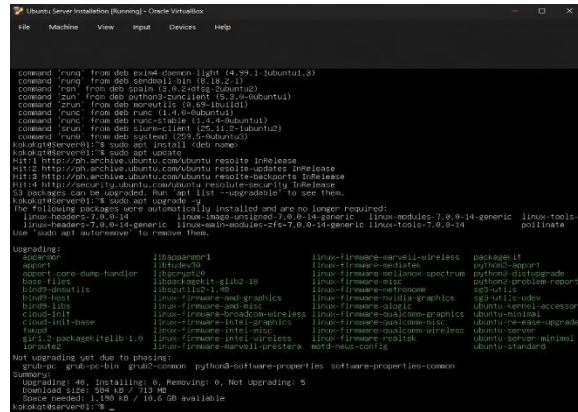
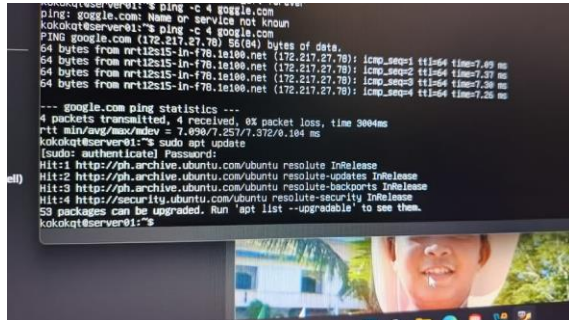


```
ping -c 4 google.com
```



Task 5 – Update the Server

```
sudo apt update sudo apt upgrade -y
```



Task 6 – Verify SSH Service

```
systemctl status ssh
```



BIOS vs UEFI Comparison

Aspect	BIOS	UEFI
Definition	Legacy firmware standard from the 1980s that initializes hardware before OS handoff.	Modern firmware standard designed to replace BIOS with more features and flexibility.
Boot process	Reads the first sector (MBR) and executes a 16-bit boot loader.	Reads .efi files directly from an EFI System Partition; supports 32/64-bit.
Max disk support	2 TB (MBR partitioning limit)	Effectively unlimited (GPT partitioning, 9.4 ZB theoretical)
Partition style	MBR, max 4 primary partitions	GPT, up to 128 partitions
Security features	None built in	Secure Boot, cryptographic verification of boot loaders
Speed	Slower boot (16-bit real mode, sequential device init)	Faster boot (32/64-bit mode, parallel init)
Advantages	Simple, broad legacy compatibility	Faster boot, larger disks, Secure Boot, GUI, network support
Disadvantages	Slow, 2TB limit, no security features, 16-bit only	More complex firmware; legacy OSes may need compatibility mode

Modern usage	Legacy systems, some older enterprise hardware	Standard on all new PCs and servers since ~2012
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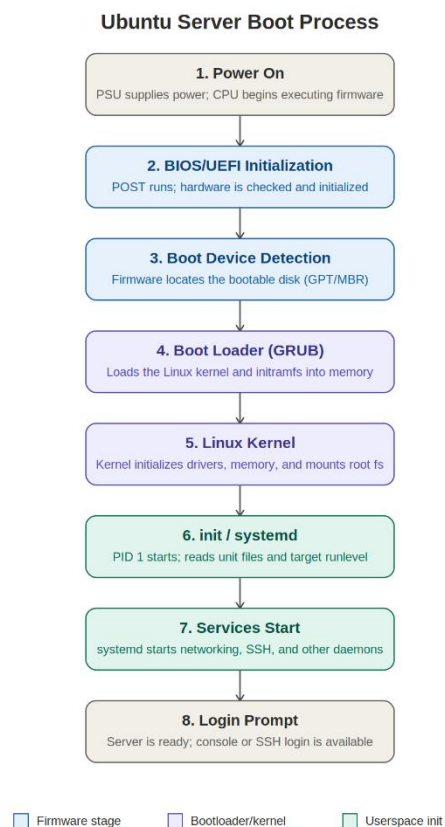
Why UEFI Has Largely Replaced BIOS

UEFI has become the standard firmware in modern computing because it directly addresses BIOS's core limitations. Its use of GPT partitioning removes the 2 TB disk-size and four-partition ceilings that constrain BIOS/MBR systems, which matters for enterprise servers with large storage arrays. UEFI also boots faster because it can initialize hardware in parallel rather than the sequential process BIOS relies on, and it operates in 32-bit or 64-bit mode instead of being limited to 16-bit real mode.

Security is another major driver of the shift. UEFI's Secure Boot feature cryptographically verifies boot loaders and OS kernels before execution, helping prevent rootkits and other boot-level malware from persisting on a system — a capability BIOS simply does not have. UEFI additionally supports richer pre-boot environments, including graphical interfaces, mouse input, and network booting, which simplify diagnostics and remote provisioning. For enterprise and cloud deployments in particular, UEFI's combination of larger disk support, faster boot, and built-in security make it the practical default, and most current operating systems — including Ubuntu Server — are built and tested primarily against UEFI firmware.

Boot Process Flowchart

The diagram below illustrates the eight stages of the Ubuntu Server boot process, from power-on through to the login prompt.



Stage summary:

- Power On – PSU energizes the system; the CPU begins executing firmware code.
- BIOS/UEFI Initialization – POST checks hardware components for faults.
- Boot Device Detection – Firmware locates the bootable disk via MBR or GPT.
- Boot Loader (GRUB) – Loads the Linux kernel and initial RAM filesystem into memory.
- Linux Kernel – Initializes device drivers, memory management, and mounts the root filesystem.
- init/systemd – PID 1 starts and reads unit files to determine the target runlevel.
- Services Start – systemd starts networking, SSH, and other configured daemons.
- Login Prompt – The system is ready for console or SSH login.

Windows Server Installation Summary

As part of the bring-home assignment, Windows Server Evaluation was installed in a separate virtual machine to compare the experience against Ubuntu Server.

Task	Status
Operating system installed	[Yes/No]
Computer name assigned	[e.g., WIN-SRV01]
Administrator password created	[Yes]
Successful login	[Yes]

OS Comparison: Windows Server vs Ubuntu Server vs Rocky Linux

Aspect	Windows Server	Ubuntu Server	Rocky Linux
Licensing	Paid, per-core/CAL licensing	Free, open source (Canonical support optional)	Free, open source (RHEL-compatible)
User interface	GUI by default, Server Core optional	Command-line by default	Command-line by default
Package management	Windows Update / MSI / winget	APT (.deb packages)	DNF/YUM (.rpm packages)
Security	Windows Defender, AD-integrated policies	AppArmor, frequent LTS security patches	SELinux, enterprise-grade hardening
Performance	Higher resource overhead, especially with GUI	Lightweight, efficient on modest hardware	Lightweight, optimized for stability
Enterprise use cases	Active Directory, .NET apps, SharePoint	Web hosting, containers, cloud-native apps	On-prem enterprise servers needing RHEL compatibility
Advantages	Deep Windows/AD ecosystem integration	Large community, frequent updates, cloud-friendly	Long-term stability, free RHEL-class OS
Disadvantages	Licensing cost, higher resource use	Shorter support cycle than RHEL family	Slower release cadence than Ubuntu

Discussion

Windows Server is the natural choice when an organization is already invested in the Microsoft ecosystem — Active Directory, Exchange, or .NET line-of-business applications — since it integrates tightly with those tools, but that convenience comes with licensing costs and a heavier resource footprint. Ubuntu Server, by contrast, is free, lightweight, and widely used for web hosting, containers, and cloud-native workloads; its large community and frequent release cycle make it a strong general-purpose choice for a startup like ABC Startup Solutions that needs to move quickly without licensing overhead.

Rocky Linux occupies a middle ground: it is free like Ubuntu but built for binary compatibility with Red Hat Enterprise Linux, making it attractive for organizations that want RHEL-class stability and a longer support lifecycle without the subscription cost. For ABC Startup Solutions' initial deployment — file sharing, web hosting, and

internal services — Ubuntu Server is the most practical fit given its low resource requirements, active community, and ease of remote administration via SSH, which matches the tasks completed in this project.

References

- Ubuntu Server Documentation – <https://ubuntu.com/server/docs>
- UEFI Specification – <https://uefi.org/specifications>
- Rocky Linux Documentation – <https://docs.rockylinux.org>
- Microsoft Windows Server Documentation – <https://learn.microsoft.com/windows-server>